Crescent Garden Cemetery Earthquake & Liquefaction Risk Elsinore Fault Scenarios M3.0 to M7.0 Probability

The Crescent Gardens Cemetery site (Meadowbrook parcels) lies 4.8 miles from the active Elsinore Fault (Public Resources Code §§ 2621–2630). Moderate-to-large earthquakes on this fault at the documented ~4.8-mile distance would produce damaging shaking and liquefaction risk. The table below shows estimated shaking at the cemetery site for M4.0 through M8.0 (based on **USGS ground-motion prediction equations for strike-slip faults and the 2023 National Seismic Hazard Model**), together with the approximate 20-year probability of one or more events of that magnitude or greater on the Elsinore Fault system (derived from **UCERF3** time-dependent forecasts).

Magnitude	Modified Mercalli Intensity (MMI) at Site	Peak Ground Acceleration (PGA)	Likely Effects at Cemetery Site	Probability in Next 20 Years (approx.)
M4.0	IV (Light)	0.01–0.05 g	Felt by many; no damage expected; negligible liquefaction risk	Very high (>99 %) – multiple events expected
M5.0	V–VI (Moderate)	0.10–0.25 g	Felt by all; light damage possible; low-to-moderate liquefaction risk	High (~50–70 %) – several events likely
M6.0	VI–VII (Strong to Very Strong)	0.25–0.50 g	Moderate damage possible; **high liquefaction risk** in perched groundwater	Moderate (~15–25 %)
M7.0	VIII–IX (Severe to Violent)	0.70–1.20 g	Major structural damage; high chance of surface rupture nearby; severe liquefaction and settlement	Low (~5–8 %) for M≥6.7+ on Elsinore
M8.0	IX–X (Violent to Extreme)	1.50+ g	Catastrophic damage; surface rupture almost certain; extreme liquefaction and permanent ground displacement	Very low (<1 %) – beyond credible maximum for this fault

U.S.G.S. Earthquake Hazards Program **Exhibit F-2**

M7.1 Scenerio Elsinore Fault

PERCEIVED SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
POTENTIAL DAMAGE	none	none	none	Very light	Light	Moderate	Mod./Heavy	Heavy	Very Heavy
PEAK ACC.(%g)	<0.05	0.3	2.8	6.2	12	22	40	75	>139
PEAK VEL.(cm/s)	*<0.02*	*0.1*	*1.4*	*4.7*	*9.6*	*20*	*41*	*86*	*>178*
INSTRUMENTAL INTENSITY	I	II–III	IV	V	VI	VII	VIII	IX	X+

Scale based upon Worden et al. (2012)

Exhibit F-3

U.S.G.S. Earthquake Hazards Program
M7.1 Scenerio Elsinore Fault - Contours

PERCEIVED SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
POTENTIAL DAMAGE	none	none	none	Very light	Light	Moderate	Mod./Heavy	Heavy	Very Heavy
PEAK ACC.(%g)	<0.05	0.3	2.8	6.2	12	22	40	75	>139
PEAK VEL.(cm/s)	*<0.02*	*0.1*	*1.4*	*4.7*	*9.6*	*20*	*41*	*86*	*>178*
INSTRUMENTAL INTENSITY	I	II–III	IV	V	VI	VII	VIII	IX	X+

Scale based upon Worden et al. (2012)